

Vignesh Pandidurai

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Data Engineer

Professional Summary

- Results-driven **Data Engineer** with **11+ years** of experience in **designing, developing, and optimizing data solutions** using **HPCC Systems, Google Cloud Platform (GCP), and Apache Airflow**.
- Skilled in **data modeling, database migrations, and automation** using **Python** and **SQL**, with expertise in handling **diverse data formats** like Parquet.
- Adept at delivering **high-performance, scalable solutions** across development and production environments, with a strong background in shipping, documentation, and logistics domains.
- Experience all phases of **Software Development Life Cycle (SDLC)** which includes **User Interaction, Business Analysis/Modeling, Development, Integration, Planning, and management of Builds**.
- Extensive hands-on experience across the entire system development lifecycle, encompassing **requirement definition, design, architecture, testing, deployment, and on-going support**.
- Proficient in **Agile** and **waterfall** methodologies, ensuring successful project execution in diverse environments.
- Proficient in **problem analysis** with strong communication and interpersonal skills and quick to grasp new concepts and technologies.
- **Extract Transform and Load** data from various source systems to Blob storage, **Ms-SQL, Google Big query SQL**.
- Worked on Projects using **HPCC data analytics cluster** computing framework.
- Worked on **Google Cloud computing, Apache airflow, DAG etc.,**
- Experience in handling automated **Python** jobs and **shell scripts**.
- Experience in working with the different file formats like Parquet.
- Extensive work experience in **Water-Fall** models building **high performance** and **high-quality** database engineering works in development and production modules
- Expertise in **Data Modelling** and **ER diagrams**.
- Expertise knowledge in **Shipping, Documentation& Logistics** file environments
- Expertise in building high performance and maintainable database objects like Tables, Indexes, Constraints, Stored procedures, Functions, Triggers and Packages.
- Developed **workflow** using **Airflow**.
- Expertise in handling filter concepts using **Synonym Schema**.
- Experience in **Database migrations, Validating** objects and Jobs
- Experience in Offsite support for newly implemented countries (Costa Rica Ecuador)
- Experience in creating email alerts based on different use cases
- Experienced in technical design and code reviews.
- Experience in **SQL Server, MySQL, and Oracle databases**.
- Highly motivated with **strong communication skills**, ability to **interact with team Members**, developers, users, and zeal to **learn new technologies**.
- Good understanding of **Spark Architecture** including **Spark Core, Spark SQL, Data Frames, Spark Streaming, Driver Node, Worker Node, Stages, Executors and Tasks**.
- Good interpersonal abilities, communication skills & maximum contribution to attain the team goal.

Technologies

- **Big Data:** Cloud Storage, Big query, Dataproc, Data transfer, Pub/Sub, Dataflow, Big query SQL.
- **Data Warehouse and Cloud Analytics:** GCP, AWS
- **Programming Languages:** Python, R, ECL, Shell Scripting
- **Cloud:** Google Cloud, AWS S3
- **Platform:** GCP
- **Database:** Oracle, SQL, PL/SQL , Google Big query
- **Markup languages:** XML, HTML
- **Platforms:** Unix, Linux
- **Version Controls:** GITHUB, GITLAB
- **Development Tools:** Visual Studio Code, SQL Developer, SSMS.
- **Collaborate tools:** Confluence,JIRA.

Certifications

- Oracle Certified Associate, SQL and PL/SQL([Certificate](#))
- Oracle Certified Java SE 6 Programmer([Certificate](#))
- Google Cloud Professional Data Engineer ([Certificate](#))

Professional Experience

ClientServer Technologies, NJ
Senior Data Engineer

Oct 2024 – Present

Responsibilities:

- Google Big Query Expertise: Proficiently created and managed Google Big Query datasets, tables, and views, tailored to support evolving project requirements and enhance data accessibility.
- Involved in Extracting Transforming and Loading data from Sources Systems to GCP/AWS Cloud Storage services using a combination of Airflow, Dataflow, Python, Pub sub and Big Query data Ingestion to one or more GCP Services and processing the data in Databricks.
- Resource Management: Effectively coordinated resource utilization and task allocation, ensuring seamless team collaboration and optimal use of available resources.
- Sprint Task Management: Oversaw and managed Data Engineering Sprint tasks, aligning them with project milestones to ensure timely and quality deliverables.
- Created Pipelines in Composer/Dataflow/Dataproc using Linked Services/Datasets/Pipeline/ to Extract, Transform, and load data from different sources like Cloud SQL, Cloud storage, Big Query Data warehouse, write-back tool and backwards.
- Developed JSON Scripts for deploying the Pipeline in dataproc that process the data using the SQL Activity and working cloud shell SDK to configure the services.
- Using G-cloud functions with python to load the data into Big Query for on arrival CSV file in GCS bucket also process and load bound and unbound data from Google PubSub topic to Big Query using Dataflow with Python.
- AWS Lambda Development: Designed and implemented AWS Lambda functions to handle server less compute tasks efficiently, enhancing system scalability and reliability.
- AWS S3 Configuration: Configured AWS S3 storage buckets to handle data ingestion from multiple sources, ensuring secure and organized data storage.
- Worked on CSV, JSON, Avro, Parquet, ORC file formats and worked on Pyspark to transform the data and optimized the long running spark jobs.
- Implemented AWS SNS and AWS SQS and used from Airflow.
- ETL Pipeline Monitoring: Conducted daily verification checks on ETL pipelines, Airflow jobs, and monitoring alerts to maintain system performance and data integrity.

- Automation for Efficiency: Developed automation scripts using Python and Shell scripting, reducing manual processing time and increasing operational efficiency.
- Log Monitoring and Analysis: Set up robust log monitoring frameworks in AWS Cloud Watch, enabling proactive identification and resolution of system issues.
- Data Pipeline Optimization: Enhanced the performance of ETL pipelines by optimizing workflows, reducing processing time, and improving reliability.
- Collaboration with Cross-Functional Teams: Partnered with cross-functional teams to align cloud, data, and automation solutions with overall business objectives, fostering a unified and results-driven approach.

Environment: Big Data, GCP Big Query, Dataproc, DataStudio, SDK, Python Client Libraries, Cloud functions, Apache beam, Cloud Storage, Cloud SQL, Dataflow, PubSub, Cloud Composer, ETL, CI/CD, Jenkins, Python, SQL, NOSQL, MSSQL, Snowflake, AWS Glue, EMR, S3, Spark,

Profost Technologies, UK
Senior Data Engineer

Jan 2024 – Sep 2024

Responsibilities:

- Played a pivotal role in new product development for content syndication from online retailer websites across multiple countries.
- Conducted daily verification checks on ETL pipelines, Airflow jobs, and monitored alerts to ensure data accuracy and system reliability.
- Created and managed Google Big Query datasets, tables, and views efficiently, tailored to project needs.
- Acted as a team coordinator, ensuring optimal resource utilization and effective task assignment.
- Oversaw and executed Sprint tasks related to Data Engineering, maintaining alignment with project timelines and objectives.
- Developed Python code to gather the data from HBase (Cornerstone) and designed the solution to implement using PySpark.
- Used PySpark, to join different tables and then used EMR to perform analytics on top of them.
- Created several Databricks Spark jobs with PySpark to perform several tables-to-table operations.
- Importing and exporting data jobs to perform operations like copying data from HDFS and to HDFS using Sqoop and developed Spark code and Spark-SQL/Streaming for faster testing and processing of data.
- Designed and implemented database structures (tables, indexes, stored procedures), achieving a 30% improvement in Oracle database efficiency.
- Managed datasets using Panda data frames and MySQL, queried MYSQL database queries from python using python-MYSQL connector and MySQL db package to retrieve information.
- Written Spark applications using Scala to interact with the PostgreSQL/Mongo database using Spark SQL Context and accessed Hive tables using Hive Context.
- Created Python scripts and tested to automate AWS S3 data upload/download and control instance operations with AWS API and instrumental in utilizing Compute Services (EC2, ELB), Storage
- Worked on multiple projects leveraging Azure cloud services defining business requirements.
- Designed dataflow pipelines to capture diagnostic logs and from blob storage and analyzed storage activity logs to have insights on the migrated data.
- Services (S3, Glacier, Block Storage, Lifecycle Management policies), Cloud Formation (JSON Templates), Elastic Beanstalk, Lambda, VPC, RDS, trusted Advisor and Cloud Watch.
- Created T-SQL procedures, functions, triggers for ETL process.
- Implemented SNS and SQS and used from Airflow.
- Created databases, schemas, tables in different formats (CSV, DELTA, PARQUET) performing data.
- Created Google custom dashboard and reports using Looker studio.
- Designed and developed custom dashboards in Looker Studio to provide real-time insights

and enhance decision- making for stakeholders.

- Participated in cost optimization by using pySpark optimization techniques and analyzed blob storage containers to check for unused datasets and moved to cool Tier, if necessary, by enabling diagnostic logs on Azure storage systems.
- Created customized reports by integrating data from multiple sources, including Google Big Query, to address specific business requirements and KPIs.
- Implemented interactive visualizations such as charts, graphs, and filters, enabling users to drill down into data for deeper analysis.
- Optimized dashboard performance by leveraging efficient data modeling and query execution, ensuring seamless user experience.

Environment: Oracle,MS Sql, PySpark, Spark, S3. JSON, ETL, SQL Server, Scala, Python, AWS, Tableau,Airflow, Snowflake,Cassandra, Map Reduce (EMR), Teradata,AWS lamda, EC2, MongoDB, AWS Glue, data bricks,Informatica,SNS

Infosys Pvt. Ltd.
Technology Lead

Jun 2022 – Jan 2024

Responsibilities:

- Redesigning Functional Architecture: Spearheaded the transformation of functional architecture by migrating legacy C++ applications to ECL (Enterprise Control Language), resulting in significant improvements in system efficiency and maintainability.
- Enhancing Data Processing: Managed and optimized complex filter logic to streamline data processing workflows, achieving a 20% improvement in speed and accuracy.
- User-Centric UI Development: Led the design and development of an intuitive, user-friendly UI for a corporate website, which enhanced overall user experience and engagement by 30%.
- Automation and Efficiency: Developed robust automation scripts using Python and Shell scripting, eliminating redundant manual processes and reducing processing time by 40%.
- Task Allocation and Team Coordination: Strategically allocated tasks based on team strengths and project needs, increasing overall team productivity by 15% while ensuring timely project deliverables.
- Mentorship and Team Development: Mentored team members to enhance technical and problem-solving skills, leading to a 25% increase in project delivery efficiency and team confidence.
- Optimizing Big Data Queries: Initiated and optimized Google Cloud Platform (GCP) services for handling big data queries, reducing query response times by 50% and improving data availability.
- Legacy System Integration: Ensured seamless integration of legacy systems during the architectural redesign, minimizing downtime and preserving critical business functions.
- Cross-Functional Collaboration: Worked closely with stakeholders across departments to align technical implementations with business goals, ensuring a high level of client satisfaction.
- Developed pySpark/MapReduce Jobs to parse the JSON or XML data.
- Performance Monitoring and Reporting: Established a framework for monitoring system performance post-migration and provided actionable insights to further optimize processes and outcomes.

Hexaware Technologies Pvt. Ltd.
Senior Software Engineer

Nov 2019 – Jun 2022

Responsibilities:

- Database Object Design and Creation: Designed and created critical database objects such as tables, indexes, constraints, stored procedures, packages, functions, and triggers, ensuring optimized performance and data integrity.
- Filter Concept Implementation: Led the design and implementation of complex filtering logic to streamline data processing, improving query efficiency and data retrieval accuracy.
- Backend Procedure Development: Designed and developed robust backend procedures to support report generation, enabling accurate and timely data analysis for business needs.
- Custom Oracle Forms and Reports: Developed and customized Oracle Forms and Reports tailored to meet specific business requirements in the Insurance domain, ensuring compliance with industry standards.
- Team Resource Allocation: Strategically allocated tasks and processes to team members based on their expertise and project priorities, enhancing productivity and meeting deadlines.
- Mentorship and Guidance: Guided team members in improving their technical skills and work approaches, fostering a collaborative and growth-oriented team environment.
- Report Automation: Automated reporting processes, reducing manual intervention and improving the timeliness of report delivery by significant margins.
- Performance Optimization: Optimized database queries and backend logic, reducing execution times and improving overall application performance.
- Data Integrity Assurance: Implemented robust data validation and integrity checks through constraints, triggers, and stored procedures, ensuring high-quality and reliable data.
- System Scalability Design: Designed database objects and structures with scalability in mind, supporting future growth in data volume and complexity.
- Cross-Team Collaboration: Collaborated with cross-functional teams, including frontend developers and business analysts, to align database development with overall project goals.
- Insurance-Specific Solutions: Tailored Oracle Forms and backend processes to address the unique needs of the Insurance industry, such as policy management and claims processing.
- Documentation and Standards Compliance: Created comprehensive documentation for database objects, ensuring adherence to best practices and facilitating knowledge sharing within the team.
- Code Review and Quality Assurance: Conducted thorough code reviews to ensure high-quality deliverables and compliance with organizational coding standards.
- Query Tuning: Enhanced the performance of complex queries through advanced optimization techniques, such as indexing and query restructuring.
- Risk Mitigation: Proactively identified and mitigated potential risks in database design and implementation, ensuring minimal downtime and system stability.
- Process Improvement: Continuously evaluated existing processes and introduced improvements to enhance efficiency and reduce operational overhead.
- Training Sessions: Conducted training sessions for team members on advanced database concepts and tools, fostering a culture of continuous learning.
- Deployment and Maintenance: Managed database deployment and maintenance activities, ensuring seamless integration and minimal disruption during updates.
- Stakeholder Communication: Maintained effective communication with stakeholders to gather requirements, provides updates, and ensures alignment with business objectives.

MSC Technologies Pt. Ltd.
Team Lead

Sep 2013 – Nov 2019

Responsibilities:

- Database Structure Optimization: Designed and implemented comprehensive database structures such as tables, indexes, and stored procedures, achieving a 30% increase in database efficiency and reducing query execution times.
- Enhanced Filtering Mechanisms: Managed and optimized filtering mechanisms, streamlining data retrieval processes and improving performance by 20% while ensuring accuracy in results.

- **Dynamic Report Generation:** Spearheaded the development of backend procedures for dynamic report generation, cutting report generation time by 25% and enabling faster decision-making.
- **Oracle Forms Development:** Designed and implemented Oracle Forms and automated reporting systems for an insurance project, enhancing the accuracy and accessibility of critical business reports.
- **Efficient Task Allocation:** Strategically allocated tasks among team members, improving workload distribution and boosting team productivity by 18%.
- **Technical Mentorship:** Mentored team members on best practices and advanced technical concepts, leading to a 15% reduction in errors and faster project delivery.
- **Data Integrity and Validation:** Established robust data validation and integrity mechanisms, ensuring consistent and reliable data across all systems.
- **Report Automation:** Automated complex reporting processes, reducing manual effort and significantly improving the timeliness and accuracy of business insights.
- **Performance Tuning:** Conducted query tuning and database optimization, minimizing resource consumption and enhancing the performance of critical business operations.
- **Collaboration with Stakeholders:** Worked closely with business stakeholders to understand requirements, translating them into effective technical solutions.
- **Cross-Functional Coordination:** Collaborated with frontend developers and business analysts to align backend systems with overall project goals and deliverables.
- **Project Risk Mitigation:** Identified potential risks in database design and implementation, implementing proactive measures to ensure system stability and uptime.
- **Documentation and Knowledge Sharing:** Developed detailed documentation for database structures and processes, facilitating knowledge sharing and future maintenance.
- **Insurance-Specific Customization:** Tailored database structures and reporting systems to address specific challenges and requirements of the insurance domain, such as policy management and claims tracking.
- **Continuous Process Improvement:** Regularly evaluated and improved existing systems and processes, ensuring they met evolving business needs and maintained optimal performance.

Technology & Environment:

Oracle 11g, SQL Developer, Oracle Forms & Reports tool.

Education Details:

BE in Computer Science and Engineering, Tamil Nadu, 2013, India